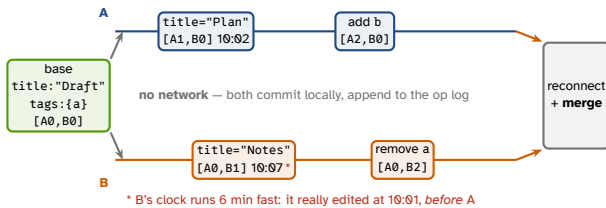


In one line: A sync engine keeps **replicas** (phones, tablets, the server) that accept writes *offline* and must end in the **same state**. Record **operations**, not snapshots, in a durable **log**; apply them **idempotently**; **detect** concurrency with **versions or vector clocks** (never wall clocks); **resolve** it with a named policy — server-authoritative, LWW, field merge, **OT** (a server orders and transforms) or **CRDTs** (merge is mathematically order-free); keep deletes as **tombstones**.

Two devices edit the same note offline



What each policy produces

policy	result
LWW, wall clock	"Notes" — the skewed clock wins; A's later edit silently lost (whole-record LWW drops A's tag b too)
server-authoritative version vectors	first to sync wins; second gets 412 / conflict , must rebase its ops [A2, B0] vs [A0, B2]: neither $\leq$ the other $\rightarrow$ concurrent $\rightarrow$ detected , not solved
3-way field merge	vs base: tags {a} $\rightarrow$ A {a,b}, B {} $\rightarrow$ {b}; title changed on <i>both</i> $\rightarrow$ ask / keep both
OT	server orders A then B, transforms B's op against A's — text editors
CRDT	title = LWW-Register on an HLC (same winner on every replica); tags = OR-Set $\rightarrow$ {b} — converge with no server

The engine

- Local store = truth**; each write also appends an **op** {opId: UUID, entity, id, field, value, baseVersion} to a durable **outbox** in the *same* transaction. Ops (intent: "add tag b") merge better than snapshots ("tags = {a,b}").
- Push**: send ops in order; the server applies each once — opId is the **idempotency key** (at-least-once delivery + idempotent apply = effectively once) and stamps a monotonic **server revision**.
- Pull**: GET /changes?since=<cursor> $\rightarrow$ {changes, cursor, hasMore}; the cursor is a server **revision**, never a timestamp. Apply + store the new cursor in **one transaction**. Cursor too old (GC'd) $\rightarrow$ **410 Gone** $\rightarrow$ full resync.
- Rebase**: remote changes arrive while local ops are pending $\rightarrow$ apply remote, replay pending ops on top (like `git rebase`) so the UI shows local intent.
- Tombstones**: a delete is a row deleted=true, version=n, synced like any change. GC only when every replica has seen it (*causal stability*) or after a TTL longer than the max offline window — else an old device **resurrects** the row.

Clocks — detection, not decoration

- Wall clock**: users and NTP move it; skew of minutes is normal. Display only.
- Lamport**: counter; on send $c+=1$, on receive $c=\max(c, \text{msg})+1$. Causal $\Rightarrow$ smaller, but a smaller number does *not* prove causality — cannot detect concurrency. Total order with a replica-id tie-break.
- Vector clock / version vector**: one counter per replica. $V1 \leq V2$ in every slot $\Rightarrow$ V1 happened-before; neither $\Rightarrow$ **concurrent**. Cost: grows with replicas (per device, not per user) — prune retired devices.
- HLC** (hybrid logical clock): (max physical seen, logical counter) — stays close to real time, monotonic even when the device clock jumps back, fits in 64 bits; a good LWW timestamp.

Example — the two smallest CRDT merges

```
struct GCounter: Hashable { // grow-only counter
  var slots: [String: Int] = [:] // replicaId -> count
  var value: Int { slots.values.reduce(0, +) }
  mutating func inc(_ r: String) { slots[r, default: 0] += 1 }
  func merged(_ o: GCounter) -> GCounter { // join = max per slot
    GCounter(slots: slots.merging(o.slots, uniquingKeysWith: max)) }
}

struct ORSet<E>: Hashable { // observed-remove set
  var adds: [E: Set<UUID>] = [:]; var removed: Set<UUID> = []
  mutating func add(_ e: E) { adds[e, default: []].insert(UUID()) }
  mutating func remove(_ e: E) { removed.formUnion(adds[e] ?? []) }
  var elements: Set<E> { // alive = a tag not removed
    Set(adds.filter { !$0.value.subtracting(removed).isEmpty }.keys) }
} // merge: union adds, union removed -> concurrent add WINS
```

Traps

- ! "Newest timestamp wins" with device clocks — skew picks the winner.
- ! Lamport order $\neq$ causality: it can't tell "after" from "concurrent".
- ! Hard delete locally — the delete never syncs; the row comes back.
- ! Counter as a plain Int "+1" replayed on retry — double count; use op ids or a G-Counter.
- ! CRDT $\neq$ correct: it converges to a state — maybe not the one users meant (invariants like "stock ≥ 0 " need a server).
- ! Cursor = "last sync time": a row committed late with an older timestamp is skipped forever.

CRDTs — why they converge

- State-based** (CvRDT): replicas ship state; merge is a **join — commutative, associative, idempotent** — so any order, any duplicates, same result. **Op-based** (CmRDT): ship ops that commute; needs causal, exactly-once delivery.
- G-Counter**: slot per replica, value = sum, merge = max per slot (a single int would double-count on re-merge). **PN-Counter** = two G-Counters ($P - N$): likes.
- LWW-Register**: (value, timestamp, replicaId), larger wins — deterministic, still *loses* an edit by design. Use an HLC, not Date().
- OR-Set**: every add gets a unique tag; remove deletes only the tags it *observed* $\rightarrow$ a concurrent add survives (**add-wins**). Tombstoned tags = metadata that grows until GC.
- Text / JSON**: sequence CRDTs (RGA, YATA) give each character an id and order — Yjs, Automerge. Cost: metadata per character, history kept.

OT vs CRDT

	OT	CRDT
idea	rewrite op B against A: $\text{ins}(5)$ after $\text{ins}(2, "xy") \rightarrow \text{ins}(7)$	data type whose merge is order-free
needs	a central server to order ops	nothing — peer-to-peer, offline for weeks
cost	transform functions, hard to prove	metadata + tombstones grow
seen in	Google Docs-style editors	local-first apps, Yjs, Automerge

Pick the policy per field, not per app

one user, many devices	field-level LWW on an HLC — rare conflicts, cheap
money · stock · bookings	server-authoritative : reject + rebase (invariants)
likes, counts	increment ops with op ids, or a PN-Counter
tags, members	OR-Set semantics (add wins)
shared text	sequence CRDT, or OT behind a server
files, blobs	keep both: "conflicted copy" + let the user pick

Apple's example — CloudKit

- NSPersistentCloudKitContainer: mirrors a Core Data store to CloudKit record zones. Model rules: **no unique constraints**, every relationship **optional + has an inverse** (records arrive out of order), no **Deny** delete rule. After promoting to production the schema is **add-only** — never rename or delete a field.
- Conflict detection: each CKRecord carries a change tag; saving a stale copy $\rightarrow$ `CKError.serverRecordChanged` (the server's record in the error).
- CKSyncEngine (iOS 17 / macOS 14): your own store; the engine **schedules** send/fetch, batches your pending changes (`nextRecordZoneChangeBatch`), and hands you a state blob to **persist**; `serverRecordChanged` is **yours** to resolve.

Remember

Ops in a log · idempotent by op id · revision cursor · versions detect · policy resolves · CRDT = merge is a join · tombstones until everyone has seen them.

Related: ios-client-system-design (sync cycle, outbox) · data-access-patterns (SSOT) · cloudkit-sync-sharing · coredata-migrations-swiftdata · distributed-system-design (CAP) · api-design (ETag, If-Match)